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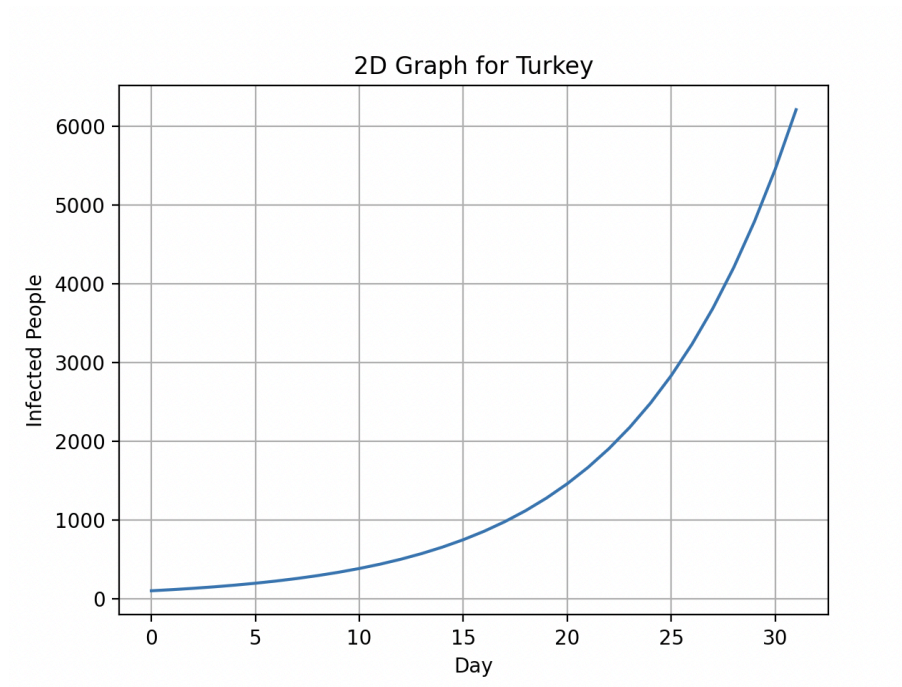
COMPE 361

Homework 1

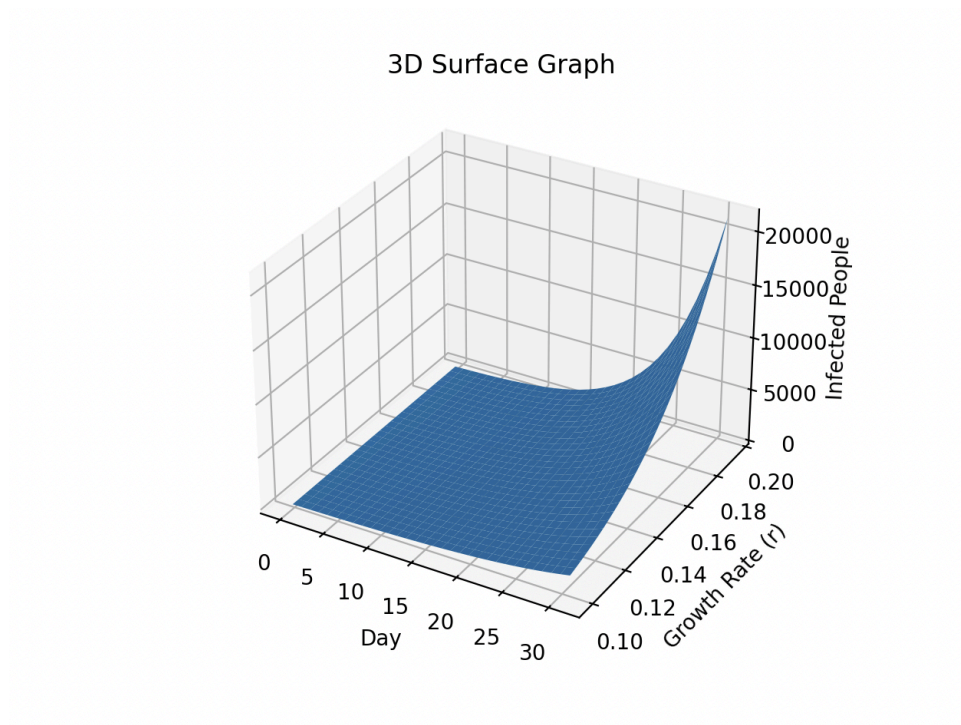
- 1) Using the logistic equation, compute the values of  $P_i$  for **31 days**.

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Turkey
Day 0: 100.00
Day 1: 114.39
Day 2: 130.85
Day 3: 149.67
Day 4: 171.20
Day 5: 195.82
Day 6: 223.98
Day 7: 256.18
Day 8: 292.99
Day 9: 335.09
Day 10: 383.23
Day 11: 438.25
Day 12: 501.15
Day 13: 573.05
Day 14: 655.21
Day 15: 749.10
Day 16: 856.37
Day 17: 978.90
Day 18: 1118.83
Day 19: 1278.59
Day 20: 1460.94
Day 21: 1669.02
Day 22: 1906.36
Day 23: 2176.96
Day 24: 2485.34
Day 25: 2836.58
Day 26: 3236.38
Day 27: 3691.14
Day 28: 4207.98
Day 29: 4794.86
Day 30: 5460.56
Day 31: 6214.77
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2) Plot a **2D graph** showing the change of ( $P_i$ ) over time for one selected country.



3) Construct a **3D surface graph** illustrating how the infection dynamics depend on the **growth rate ( $r$ )**:



- 4) As the day number  $i$  increases, the number of infected people  $P_i$  grows rapidly at first because the infection spreads quickly when few people are infected. After some time, the growth begins to slow down as more of the population becomes infected. Eventually, the increase becomes very small and the curve levels off, approaching the maximum value  $N$ .
- 5) The number of infected people approaches  $N$  because the growth term includes the factor  $(1 - P_i/N)$ . When  $P_i$  is small, this term is close to 1, so the infection grows quickly. As  $P_i$  gets closer to  $N$ , the term becomes smaller, which slows the growth. Eventually, when  $P_i$  is near  $N$ , the growth becomes almost zero, so the number of infected people levels off and stabilizes around  $N$ .